Data Sheet

EXPERIMENT NO. 2(a)

Title: - Study of Ripple Factor and Voltage Regulation Characteristics of Full-Wave Rectifier Without Capacitor Filter (using Tinkercad software circuit).

Apparatus Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	Oscilloscope	virtual instrument	1
2.	Centre Tapped Transformer	230v – 12v	1
3	Resistor	10KΩ, 50KΩ, 100KΩ	1
4.	Diode	IN4007	2
5.	Ground	Terminal mode	1
6.	Signal Generator	Generator	1

Important Formulae: -

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$V_{ac} = \sqrt{(V_{rms}^2 - V_{dc}^2)}$$

$$V_{dc} = \frac{2V_m}{\pi}$$

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Title :- Study of Ripple factor & Voltage Regulation characteristics of Full Wave Rectification without Capacitor filter.

Theory :- A full wave rectifier convert both half cycle of Ac input signal into a pulsating DC output. It generally uses or a bridge with a center-tapped transformer or a bridge of four diodes. In this experiment the output contains a considerable AC component called Ripple.

» Ripple factor is the Ratio of the RMS value of AC Component to the DC component of the rectifier output

$$\gamma = \frac{Ac(rms)}{V_{DC}}$$

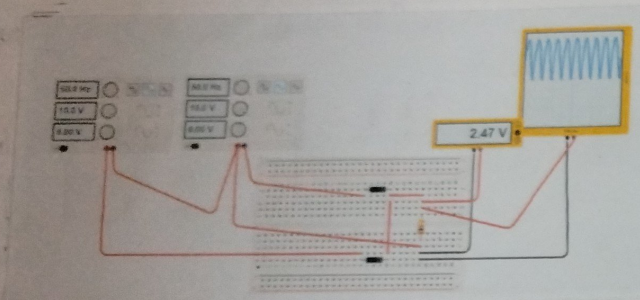
Component Used :-

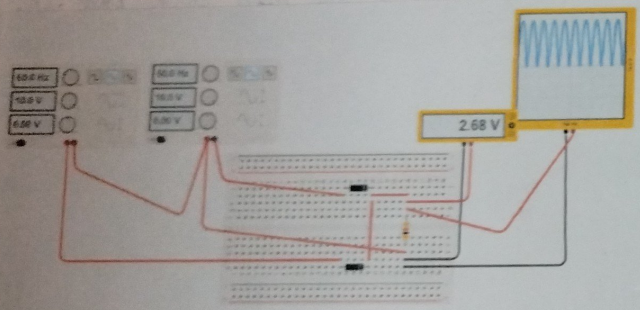
SL.NO	Apparatus	Specifications	Quantity
1.	Ac Voltage Source	12v, 50Hz	1
2.	Step down Transformer	230v / 12v	1
3.	Diode	1N4007	4
4.	load Resistor	1Ω	1
5.	Multimeter	Digital	1
6.	Oscilloscope	2-Channel	1
7.	Connecting Wires	-	As required
8.	Breadboard	Standard	1

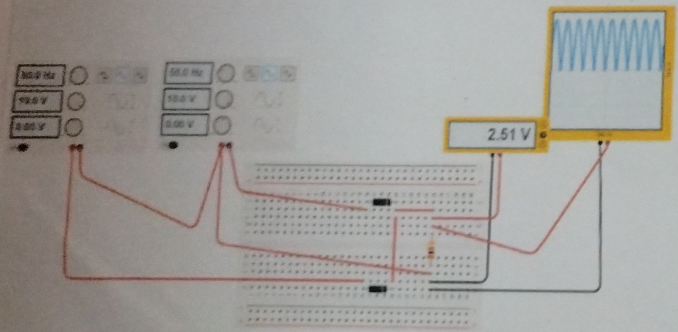
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Observation Table:-

Without Capacitor filter:-

RL	V_{dc} (V)	V_{ac} (V)	R.F (V)	R.F (V)
1 KΩ	6.20	3.05	0.49	0.48
2 KΩ	6.35	3.08	0.49	0.48
3 KΩ	6.45	3.10	0.49	0.48

Voltage Regulation:-

Condition	Output Voltage
No load (VNL)	6.60 V
Full load (VFL)	6.20 V

#Calculation:-

$V_{rms} = V_m/\sqrt{2}$ =

$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2}$

$V_{dc} = 2V_m/\pi$

Ripple factor $= \sqrt{(V_{rms}/V_{dc})^2 - 1} = 0.48$.

(Theory)

Ripple factor (Prac) $= \frac{V_{ac}}{V_{dc}} = 0.48$

Voltage Regulation $= [(V_{NL} - V_{FL})/V_{FL}] \times 100$

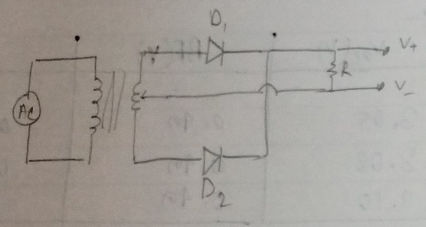
$= (6.60 - 6.20) \times 100 / 6.20$

$= 6.45\%$.

~~For 1KΩ → $V_{rms} = V_m/\sqrt{2}$ =~~

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Circuit Diagram:-



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Conclusion:-

The full wave rectifier Converts both half cycle of AC input into a pulsating DC output. The ripple factor & voltage Regulation were studied Succesfully without using a Capacitor filter. The obtained result were found to be approximately close to the theoritical values verifing the Working of the full-wave rectifier.

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Data Sheet

EXPERIMENT NO. 2(b)

Title: - Study of Ripple Factor and Voltage Regulation Charac
Wave Rectifier With Capacitor Filter (using Tinkercad software

Apparatus Required: -

Sl. No.	Apparatus / Component	Specification	
1.	Signal Generator	Generator	
2.	Oscilloscope	Virtual Instrument	
3.	Center Tapped Transformer	Trans - 2 0 P35	
4.	Resistor		
5.	Capacitors		
6.	Diode	IN4007	
7.	Connecting Wires		

Important Formulae: -

Ripple factor (Theoretical) $\gamma = \frac{1}{4\sqrt{3}fCR}$

$V_{ac} = \frac{V_{r(p-p)}}{2\sqrt{3}}$

$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$

Ripple Factor $\gamma = \frac{V_{ac}}{V_{dc}}$

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Title: Study of Ripple factor & voltage Regulation characteristics of full wave rectifier with Capacitor filter.

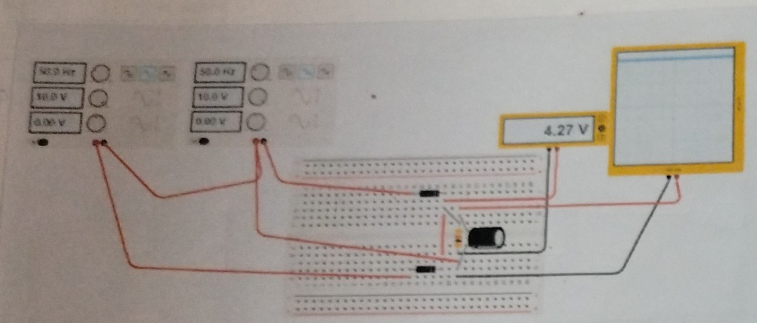
Theory:- Full-wave rectifier with a capacitor filter converts Ac into a smoother DC voltage by utilizing both half cycle of the input AC. It charges to the peak voltage. When the rectified voltage when the rectified voltage rises & discharges through load. The ripple factor indicates the amount of AC components present in the output. A lower ripple factor means a smoother DC output.

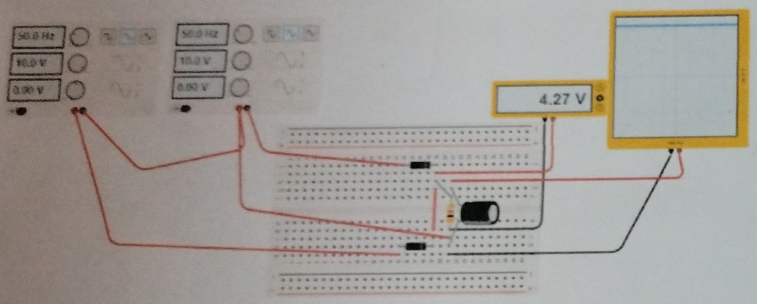
Components Used:-

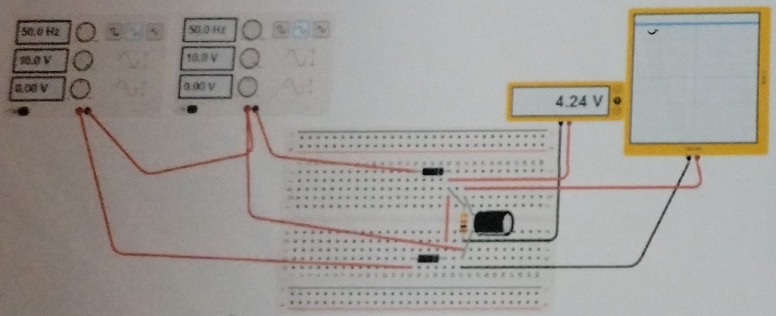
Sl No.	Apparatus	Specification	Quantity
1.	AC voltage Source	230V 50Hz	1
2.	Step down Transformer	230V / 12V	1
3.	Diode	IN 4007	4
4.	Capacitor	100 μf, 25V	1
5.	load Resistor	1KΩ	1
6.	Digital Multimeter	DC/AC voltage	1
7.	Oscilloscope	2-chanel	1
8.	Breadboard	Standard	1
9.	Connecting Wires.	—	10

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Observation Table :-

Without Capacitor Filter

C (μF)	R_L (kΩ)	V_{dc} (V)	V_{ac} (V)	Ripple factor (practical)	Ripple factor (Theo)
100	1	14.82	0.43	0.29	0.029
100	2	15.10	0.22	0.19	0.019
100	3	15.21	0.15	0.010	0.010

Voltage Regulation :-

Condition	Output voltage
No load (VNL)	15.50 V
Full load (VFL)	14.82 V

Calculation :-

Ripple factor = $\gamma = \frac{1}{4\sqrt{3}fCR}$

$V_{ac} = V_r(p\text{-}p)/2\sqrt{3}$

$V_{dc} = V_m - V_m(p\text{-}p)/2$

Ripple factor = V_{ac}/V_{dc}

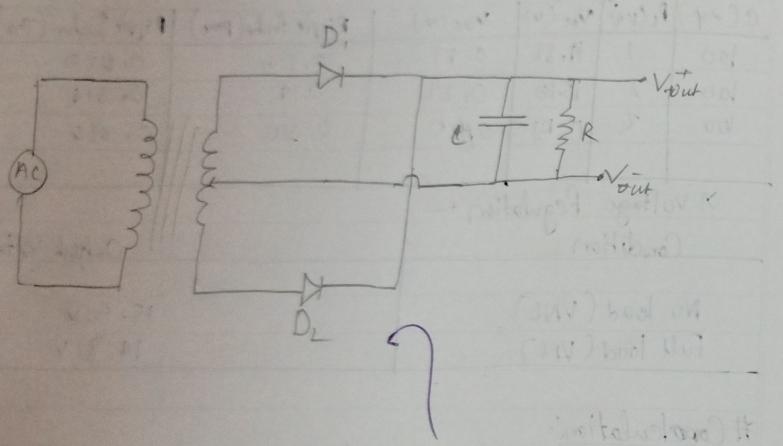
Voltage Regulation = $[(V_m - V_{FL})/V_{FL}] \times 100 = \frac{(15.50 - 14.82)}{14.82} \times 100$

$= 4.859\ \%$

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Circuit Diagram:-



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Conclusion:- The experiment successfully demostrated the Ripple factor & voltage Regulation characteristics of a full wave rectifier with a capacitor filter. The capacitor filter effectively reduced the ripple & produced a smoother DC output. It was also observed that the output voltage decrease slightly as the load increase.

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